

Muhammad Ahmad

Full Stack | AI / ML Engineer

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with over 3 years of professional experience designing and deploying intelligent systems across computer vision, machine learning, and large language model (LLM) architectures. Proven track-record building production-grade RAG pipelines, agentic reasoning workflows (Chain-of-Thought, ReAct), and real-time voice-enabled AI services using Python and FastAPI. Hands-on expertise with deep learning model training (CNNs, YOLO), cloud deployment on AWS, and vector database integration. Published researcher in AI-based cybersecurity.

SKILLS

Programming Languages: Python, JavaScript, SQL

Frameworks & Libraries: FastAPI, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, PyMuPDF, Electron.js, Bootstrap, Ruby on Rails

AI & Machine Learning: Large Language Models (LLMs), RAG Pipelines, Chain-of-Thought (CoT), ReAct, CNNs, YOLO (v8–v11), Computer Vision, NLP

Databases & Vector Stores: PostgreSQL, MySQL, ChromaDB, FAISS

Cloud & DevOps: AWS (EC2, ECR, ECS, S3, Cognito), Azure OpenAI, Docker

APIs & Tools: GPT-4 Realtime API, Whisper ASR, Groq LLM API, IPC, Git

WORK EXPERIENCE

Careonix

January 2026 – Present

Full Stack AI Engineer

Lahore, Pakistan

- Design and deploy agentic AI systems leveraging LLM reasoning, tool-calling, and retrieval-augmented generation (RAG) to automate complex clinical workflows.
- Build real-time, voice-enabled AI pipelines using Azure OpenAI (GPT-4 Realtime, Whisper ASR) and FastAPI streaming APIs, reducing average query response latency by over 40%.
- Develop and maintain cloud-native AI services on AWS (ECS, EC2, S3) with Docker containerization and Cognito-based authentication, achieving 99.9% uptime in production.
- Integrate PostgreSQL and vector databases to power scalable semantic search and context retrieval across large document corpora.

Cyberautix Technologies

March 2023 – December 2025

Software Engineer

Lahore, Pakistan

- Architected and shipped AI-powered desktop applications using Electron.js, bridging FastAPI Python inference services via IPC for secure, low-latency decision-making and workflow automation.
- Built end-to-end computer vision and document-processing pipelines using TensorFlow, YOLO (v8–v11), OpenCV, and PyMuPDF — covering dataset curation, model training, evaluation, and productionization across 5+ client projects.
- Optimized image-processing pipelines with NumPy vectorization and OpenCV, improving model inference speed by approximately 30% and increasing detection accuracy by 15%.
- Delivered 3+ production-ready desktop AI applications, each integrating custom-trained models via REST APIs, directly enhancing client operational efficiency.

Devsinc

December 2022 – February 2023

Software Engineer Intern

Lahore, Pakistan

- Developed and maintained Ruby on Rails web application features, contributing to a codebase serving thousands of active users.
- Implemented responsive UI components using Bootstrap, improving cross-device compatibility across the product.
- Participated in daily code reviews and Agile sprint ceremonies, adopting professional engineering practices under senior mentor guidance.

KEY PROJECTS

CareOnix | FastAPI, PostgreSQL, pgvector, RAG, LLMs, AWS

- Built a production multi tenant AI knowledge platform enabling organizations to securely upload internal documents and interact with an AI assistant powered by Retrieval Augmented Generation and vector search.

AI Voice Receptionist for Dental Clinics | FastAPI, Azure OpenAI, Whisper ASR, AWS, RAG

- Developed an AI powered voice receptionist capable of understanding caller intent, managing appointment workflows, answering clinic specific questions using Retrieval Augmented Generation, and automating patient communication.

InsightLens | Python, FastAPI, ChromaDB, RAG, LLMs

- Designed and developed an AI-powered insight-generation platform integrating RAG pipelines with LLM reasoning, enabling users to query large document collections with high accuracy.

CricketSense | Python, FAISS, Groq LLM API, CoT, ReAct

- Built a semantic cricket reasoning assistant using FAISS vector retrieval and Chain-of-Thought (CoT) prompting with ReAct reasoning, achieving contextually accurate responses on complex sports queries.

Musculoskeletal Radiograph Abnormality Detection System | TensorFlow, CNNs

- Developed a final-year capstone integrating multiple CNN models into a production-ready desktop application for automated medical image classification and abnormality detection.

Published Research | Springer — "Classification of Cloud Attacks Using Deep Learning"

- Applied deep learning models to classify cloud network traffic for cybersecurity threat detection; research accepted and published in Springer academic journal.

EDUCATION

University of Central Punjab

2019 – 2023

Bachelor of Science in Computer Science (BSCS)

Lahore, Pakistan

CGPA: 3.76 / 4.0

Superior College, Multan Road Campus

2017 – 2019

Intermediate in Computer Science (ICS)

Lahore, Pakistan

CERTIFICATIONS

- Software Engineer-HackerRank
- RAG Course for Beginners – Simplilearn
- Artificial Intelligence Course – Samsung Innovation Campus
- Ruby on Rails Bootcamp – Devsinc
- Programming Foundations with JavaScript, HTML, and CSS – Coursera